

Lab Experiment 1

Q1.) Implement C program to display a message "Hello World".

INPUT→

```
C Assignment_3.c > main()
1  #include<stdio.h>
2
3  int main(){
4      printf("Name: ADITYA RAJ, Roll_no: 25/B19/084\n");
5      printf("Hello World");
6      return 0;
7  }
```

OUTPUT→

```
PS C:\C_Demos\C Demos> cd "c:\C_Demos\C Demos"
Name: ADITYA RAJ, Roll_no: 25/B19/084
Hello World
PS C:\C_Demos\C Demos>
```

Q2.) Implement C Program to illustrate Arithmetic operators. This program takes an integer variable 1 and performs the basic arithmetic functions like Addition, Subtraction, Multiplication, Division, Unary Addition and Unary Subtraction.

INPUT→

```
C Assignment_3.c > main()
1  #include<stdio.h>
2  int main(){
3      printf("Name: ADITYA RAJ, Roll_No: 25/B19/084\n");
4      int var1;
5      printf("Enter the number: \n");
6      scanf("%d",&var1);
7      char operator;
8      printf("Enter the operator to perform on operand:\n");
9      scanf(" %c",&operator);
10     if(operator=='+'){
11         printf("The addition of %d with itself is %d", var1, var1 + var1);
12         printf("\nThe unary addition of %d is %d only", var1, var1);
13     }else if( operator=='-'){
14         printf("The subtraction of %d with itself is %d", var1, var1-var1);
15         printf("\nThe unary subtraction of %d is %d", var1, -1*var1);
16     }else if(operator=='*'){
17         printf("The Multiplication of %d with itself is %d", var1, var1*var1);
18     }
19     }else if(operator=='/'){
20         if(var1!=0){
21             printf("The Division of %d with itself is %d", var1, 1);
22         }else{
23             printf("Division of zero not defined!");
24         }
25     }else{
26         printf("INVALID OPERATOR!");
27     }
28     return 0;
29 }
```

OUTPUT→

```

PS C:\C_Demos\C_Demos> cd "c:\C_Demos\C_Demos"
Name: ADITYA RAJ, Roll_No: 25/B19/084
Enter the number:
10
Enter the operator to perform on operand:
+
The addition of 10 with itself is 20
The unary addition of 10 is 10 only
PS C:\C_Demos\C_Demos> cd "c:\C_Demos\C_Demos"
Name: ADITYA RAJ, Roll_No: 25/B19/084
Enter the number:
-16
Enter the operator to perform on operand:
-
The subtraction of -16 with itself is 0
The unary subtraction of -16 is 16
PS C:\C_Demos\C_Demos> cd "c:\C_Demos\C_Demos"
Name: ADITYA RAJ, Roll_No: 25/B19/084
Enter the number:
13
Enter the operator to perform on operand:
*
The Multiplication of 13 with itself is 169
PS C:\C_Demos\C_Demos> cd "c:\C_Demos\C_Demos"
Name: ADITYA RAJ, Roll_No: 25/B19/084
Enter the number:
21
Enter the operator to perform on operand:
/
The Division of 21 with itself is 1
PS C:\C_Demos\C_Demos> cd "c:\C_Demos\C_Demos"
Name: ADITYA RAJ, Roll_No: 25/B19/084
Enter the number:
0
Enter the operator to perform on operand:
/
Division of zero not defined!
PS C:\C_Demos\C_Demos> cd "c:\C_Demos\C_Demos"
Name: ADITYA RAJ, Roll_No: 25/B19/084
Enter the number:
54
Enter the operator to perform on operand:
%
INVALID OPERATOR!
PS C:\C_Demos\C_Demos>

```

Q3.) Implement C program which demonstrates the working of increment (++) and decrement (--) operators. Increment operator ++ adds 1 to its operand and Decrement operator -- subtracts 1 from its operand.

INPUT→

```

C Assignment_3.c > main()

1  #include<stdio.h>
2
3  int main(){
4      printf("Name: ADITYA RAJ, Roll_No: 25/B19/084\n");
5      //Post-Increment Operator(i++)
6
7      int x;
8      printf("Enter the value of x: \n");
9      scanf("%d",&x);
10     int y = x++;
11
12     //Pre-Increment Operator(++i)
13     int z = ++x;
14
15     printf("Values of x,y and z are: %d, %d and %d\n",x,y,z);
16
17     //Post-Decrement Operator(i--)
18     int u;
19     printf("Enter the value of u: \n");
20     scanf("%d",&u);
21     int v = u--;
22
23     //Pre-Decrement Operator(--i)
24     int w = --u;
25
26     printf("Values of u,v and w are: %d, %d and %d",u,v,w);
27     return 0;
28 }

```

OUTPUT→

```

Name: ADITYA RAJ, Roll_No: 25/B19/084
Enter the value of x:
12
Values of x,y and z are: 14, 12 and 14
Enter the value of u:
15
Values of u,v and w are: 13, 15 and 13
PS C:\C_Demos\C Demos>

```

Q4.) Implement C program to find the area of a circle, given the radius.

INPUT→

```

C Assignment_3.c > main()

1  #include<stdio.h>
2
3  int main(){
4      printf("Name: ADITYA RAJ, Roll_NO: 25/B19/084\n");
5
6      int radius, area;
7      float pi = 3.14;
8
9      printf("Enter the radius of the circle:\n");
10     scanf("%d",&radius);
11
12     area = pi* radius*radius;
13     printf("The Area of the circle is: %d", area);
14
15     return 0;
16 }

```

OUTPUT→

```

PS C:\C_Demos\C Demos> cd "c:\C_Demos\C Demos"
Name: ADITYA RAJ, Roll_NO: 25/B19/084
Enter the radius of the circle:
3
The Area of the circle is: 28
PS C:\C_Demos\C Demos>

```

Q5.) Implement C program to calculate the area of: (a) Trapezium

INPUT→

```

C Assignment_3.c > main()

1  #include<stdio.h>
2
3  int main(){
4      printf("Name: ADITYA RAJ, Roll_No: 25/B19/084\n");
5      int height, side1, side2, area;
6      printf("Enter the height of the trapezium:\n");
7      scanf("%d",&height);
8
9      printf("Enter the length of the first parallel side:\n");
10     scanf("%d",&side1);
11
12     printf("Enter the length of the second parallel side:\n");
13     scanf("%d",&side2);
14     area = (side1 + side2) * height / 2;
15
16     printf("The area of the trapezium is: %d\n", area);
17     return 0;
18 }

```

OUTPUT→

```
PS C:\C_Demos\C Demos> cd "c:\C_Demos\C Demos"
Name: ADITYA RAJ, Roll_No: 25/B19/084
Enter the height of the trapezium:
12
Enter the length of the first parallel side:
20
Enter the length of the second parallel side:
10
The area of the trapezium is: 180
PS C:\C_Demos\C Demos>
```

(b.) Rhombus

INPUT→

```
C Assignment_3.c > main()
1  #include<stdio.h>
2
3  int main(){
4      printf("Name: ADITYA RAJ, Roll_No: 25/B19/084\n");
5      int diagonal_1, diagonal_2, area;
6      printf("Enter the length of the first diagonal:\n");
7      scanf("%d",&diagonal_1);
8      printf("Enter the length of the second diagonal:\n");
9      scanf("%d",&diagonal_2);
10     area = (diagonal_1 * diagonal_2) / 2;
11     printf("The area of the rhombus is: %d\n", area);
12     return 0;
13 }
```

OUTPUT→

```
Name: ADITYA RAJ, Roll_No: 25/B19/084
Enter the length of the first diagonal:
12
Enter the length of the second diagonal:
15
The area of the rhombus is: 90
PS C:\C_Demos\C Demos>
```

(c.) Parallelogram

INPUT→

```
C Assignment_3.c > main()
1  #include<stdio.h>
2
3  int main(){
4      printf("Name: ADITYA RAJ, Roll_No: 25/B19/084\n");
5      int Base, Height, area;
6      printf("Enter the length of the Base of the Parallelogram:\n");
7      scanf("%d",&Base);
8      printf("Enter the Height of the parallelogram:\n");
9      scanf("%d",&Height);
10     area = (Base * Height) ;
11     printf("The area of the rhombus is: %d\n", area);
12     return 0;
13 }
```

OUTPUT→

```
PS C:\C_Demos\C Demos> cd "c:\C_Demos\C Demos"
Name: ADITYA RAJ, Roll_No: 25/B19/084
Enter the length of the Base of the Parallelogram:
25
Enter the Height of the parallelogram:
15
The area of the rhombus is: 375
PS C:\C_Demos\C Demos>
```

(d.) Cube

INPUT→

```
C Assignment_3.c > main()
1  #include<stdio.h>
2
3  int main(){
4      printf("Name: ADITYA RAJ, Roll_No: 25/B19/084\n");
5      int side, area;
6      printf("Enter the side of cube:\n");
7      scanf("%d",&side);
8      area = side*side*side;
9      printf("The area of the cube is: %d\n", area);
10 return 0;
11 }
```

OUTPUT→

```
PS C:\C_Demos\C Demos> cd "c:\C_Demos\C Demos"
Name: ADITYA RAJ, Roll_No: 25/B19/084
Enter the side of cube:
9
The area of the cube is: 729
PS C:\C_Demos\C Demos>
```

Q6.) Implement C program to calculate the surface area, volume, and space diagonal of cuboids.

INPUT→

```

C Assignment_3.c > main()

1  #include <stdio.h>
2  #include <math.h>
3
4  int main(){
5      printf("Name: Aditya Raj, Roll_No: 25/B19/084\n");
6      int length, breadth, height, area, volume, diagonal;
7
8      printf("Enter the length of the cuboid : \n");
9      printf("Enter the breadth of the cuboid : \n");
10     printf("Enter the height of the cuboid : \n");
11     scanf("%d %d %d", &length, &breadth, &height);
12
13     area = 2*(length*breadth + breadth*height + height*length);
14     volume = length*breadth*height;
15     diagonal = sqrt(length*length + breadth*breadth + height*height);
16
17     printf("The surface area of the cuboid is : %d\n", area);
18     printf("The volume of the cuboid is : %d\n", volume);
19     printf("The diagonal of the cuboid is : %d\n", diagonal);
20
21     return 0;
22 }

```

OUTPUT→

```

PS C:\C_Demos\C Demos> cd "c:\C_Demos\C Demos"
Name: Aditya Raj, Roll_No: 25/B19/084
Enter the length of the cuboid :
Enter the breadth of the cuboid :
Enter the height of the cuboid :
12
14
10
The surface area of the cuboid is : 856
The volume of the cuboid is : 1680
The diagonal of the cuboid is : 20
PS C:\C_Demos\C Demos>

```

Q7.) Implement C program to calculate the surface area, volume of Cone.

INPUT→

```

C Assignment_3.c > main()

1  #include <stdio.h>
2  #include <math.h>
3
4  int main() {
5      printf("Name: Aditya Raj, Roll_No: 25/B19/084\n");
6
7      int radius, height;
8      float area, volume;
9
10     printf("Enter the radius of the cone : \n");
11     scanf("%d", &radius);
12
13     printf("Enter the height of the cone : \n");
14     scanf("%d", &height);
15
16     area = 3.14 * radius * (radius + sqrt((height * height) + (radius * radius)));
17     volume = (3.14 * radius * radius * height) / 3.0;
18
19     printf("The area of the cone is : %.2f\n", area);
20     printf("The volume of the cone is : %.2f\n", volume);
21
22     return 0;
23 }

```

OUTPUT→

```

PS C:\C_Demos\C Demos> cd "c:\C_Demos\C Demos"
Name: Aditya Raj, Roll_No: 25/B19/084
Enter the radius of the cone :
10
Enter the height of the cone :
16.91
The area of the cone is : 906.45
The volume of the cone is : 1674.67
PS C:\C_Demos\C Demos>

```

Q8.) Implement C program to calculate the surface area, volume of the Sphere.

INPUT→


```

C Assignment_3.c > main()

1  #include <stdio.h>
2  #include <math.h>
3
4  int main() {
5      printf("Name:Aditya Raj, Roll_No: 25/B19/084\n");
6
7      int radius;
8      float area, volume;
9
10     printf("Enter the radius of the sphere:\n");
11     scanf("%d", &radius);
12
13     area = 4 * 3.14 * pow(radius, 2);
14     volume = (4.0 / 3.0) * 3.14 * pow(radius, 3);
15
16     printf("The surface area of the sphere is: %.2f\n", area);
17     printf("The volume of the sphere is: %.2f\n", volume);
18
19     return 0;
20 }

```

OUTPUT→

```

Name:Aditya Raj, Roll_No: 25/B19/084
Enter the radius of the sphere:
1.96
The surface area of the sphere is: 12.56
The volume of the sphere is: 4.19
PS C:\C_Demos\C Demos>

```

Q9.) Implement C Program to compute the area of an isosceles triangle.

INPUT→

```

C Assignment_3.c > main()

1  #include <stdio.h>
2  #include <math.h>
3
4  int main() {
5      printf("Name: Aditya Raj, Roll_No: 25/B19/084\n");
6
7      int repeat_side, side_2;
8      float area;
9
10     printf("Enter the length of the two equal sides of the isocles triangle : \n");
11     scanf("%d", &repeat_side);
12
13     printf("Enter the length of the third side of the isocles triangle : \n");
14     scanf("%d", &side_2);
15
16     area = (side_2 / 4.0) * sqrt((4.0 * repeat_side * repeat_side) - (side_2 * side_2));
17
18     printf("The area of the isocles triangle is : %.2f\n", area);
19
20     return 0;
21 }

```

OUTPUT→

```
Name: Aditya Raj, Roll_No: 25/B19/084
Enter the length of the two equal sides of the isosceles triangle :
40
Enter the length of the third side of the isosceles triangle :
50
The area of the isosceles triangle is : 780.62
PS C:\C_Demos\C Demos>
```

Q10.) Implement C program to calculate Simple Interest.

INPUT→

```
C Assignment_3.c > main()
1  #include <stdio.h>
2
3  int main() {
4      printf("Name: Aditya Raj, Roll_No: 25/B19/084\n");
5
6      float principal, rate, time, si;
7
8      printf("Enter the Principal amount: ");
9      scanf("%f", &principal);
10
11     printf("Enter the Rate of interest: ");
12     scanf("%f", &rate);
13
14     printf("Enter the Time period (in years): ");
15     scanf("%f", &time);
16
17     si = (principal * rate * time) / 100.0;
18
19     printf("The Simple Interest is: %.2f\n", si);
20     printf("The Total Amount is: %.2f\n", principal + si);
21
22     return 0;
23 }
```

OUTPUT→

```
PS C:\C_Demos\C Demos> cd "c:\C_Demos\C Demos"
Name: Aditya Raj, Roll_No: 25/B19/084
Enter the Principal amount: 25675
Enter the Rate of interest: 8.34
Enter the Time period (in years): 24
The Simple Interest is: 51391.08
The Total Amount is: 77066.08
PS C:\C_Demos\C Demos>
```